

Superstore Retail Sales Exploratory Data Analysis

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Data Analytics Portfolio

Python | pandas | NumPy | Matplotlib | Seaborn | Jupyter Notebook

\$145,454

Technology Profit

-0.86

Discount / Margin Correlation

\$725,457

West Region (Top)

Nov

Peak Sales Month

Profitability by Category

Technology: \$145,454 profit — highest performing category

Office Supplies: \$122,491 — solid mid-tier performer

Furniture: \$18,451 — significantly underperforms

Loss-making sub-categories: Tables (-\$17,725), Bookcases (-\$3,472), Supplies (-\$1,189)

Top sub-category: Copiers at \$55,617 profit

Discounting Impact

Discount vs Profit Margin correlation: **-0.86** — the strongest relationship in the dataset

Heavy discounting is the single biggest threat to profitability

High sales orders with large discounts frequently result in net losses

Sales vs Profit

Sales and Profit correlation: **+0.48** — moderate, not guaranteed

The largest sale in the dataset (~\$22,500) generated a **loss of ~-\$2,000**

Several \$5K–\$8K orders also resulted in significant losses

High sales volume alone is not a reliable profitability indicator

Regional Performance

West: \$725,457 — top performing region

East: \$678,781 — second highest

Central: \$501,240 — mid performer

South: \$391,721 — lowest, significant gap vs West

All regions contain outliers; South has the most extreme single outlier

Seasonal Trends

Peak month: November (~\$352,000) — holiday & year-end driven

Lowest month: February (~\$60,000)

Second half of year consistently outperforms first half

Unexpected March spike (~\$205,000) — likely Q1 business purchasing

October dip before Q4 surge is a recurring pattern

Profit Distribution

Majority of orders generate profit close to zero

Slight right skew — positive tail extends to ~\$8,500

Negative tail reaches ~-\$7,000

Distribution confirms most transactions are low-margin

Business Recommendations

1	Enforce stricter discount limits The -0.86 correlation between discount and profit margin shows discounting is the biggest lever to address immediately.
2	Double down on Technology Highest profit category at \$145K. Prioritise Copiers and Phones which are the top sub-categories.
3	Review loss-making sub-categories Tables, Bookcases and Supplies are all generating negative profit. Evaluate repricing or discontinuation.
4	Replicate West/East strategy in the South South trails West by \$333K in sales. Investigate what drives top region performance and apply learnings.
5	Off-season marketing in Jan–Feb Targeted promotions during the two lowest sales months could reduce seasonal revenue volatility.